

SE Lab 3 – Architectural Justification

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Section: F

System: Disaster Relief Supply & Volunteer Coordinator

Architecture Selection

We chose Layered Architecture for the Disaster Relief Supply & Volunteer Coordinator System.

Reason 1: Offline support is needed across the whole system, not in one feature

NFR-002 requires the app to queue requisition and delivery status updates made while a device is offline and sync them within 60 seconds of reconnecting, without losing or duplicating any of them. In a layered design this is solved once. The Offline Sync & Cache and Data Repository components in the Data Layer own the write queue, the duplicate checks and the encrypted cache. The business components simply call IRequisitionSync, IStatusSync and IStockLedger, and do not need to know whether a write went to the server or to the local queue. With microservices the same queue and replay logic would have to be repeated inside every service.

Reason 2: Allocating supplies must happen as a single transaction

In UC-02 the system checks warehouse stock, computes the delivery priority, reduces the stock, creates the logistics manifest and writes the audit entry. All of these must succeed together or none of them should happen, and alternate flow 4a5 says no stock may be reserved if it fails. Because the Allocation Manager and the Data Repository are layers of one application, this is a single local database transaction. If these steps were separate microservices we would need a distributed transaction with compensation logic, running over a network that is already unreliable. Stock reduced twice means supplies a shelter is expecting never arrive.

Security Advantage

Every read and write goes through one component, the Data Repository, and the Presentation Layer cannot reach storage directly. Because there is only one entry point, the security rules are applied in a single place. AES-256 encryption of the cached GIS data and the queued requisitions is done there, as NFR-001 requires. The role check from UC-02, where only a Disaster Manager with the "Allocation" role may allocate supplies, is enforced there as well, along with the audit log. If a field device is lost or a client is tampered with, it can still only call Business Layer interfaces, so it cannot read the stored data directly or skip the permission check.

Performance Benefit

The Mapping & GIS component looks in the local tile cache first through ITileCache and only calls the external GIS Data Service when a tile is missing. The mapping dashboard can therefore draw the last synced layers with no network request at all, which is how it stays within the 500 KB transfer and 3 second render limits in NFR-001 on a slow connection. Calls between layers are also ordinary function calls inside one application instead of HTTP requests between services, so a single allocation in UC-02 costs one local transaction instead of four or five network round trips.

Why not the other styles

Microservices were rejected because their main benefits, independent scaling and independent deployment, solve problems this system does not have, while their costs of network latency and eventual consistency fall on the network, which is the least reliable part of a disaster response. A plain Client-Server design with a thin client was rejected because the server is a single point of failure and the client would stop working whenever the connection drops, which breaks NFR-001 and NFR-002.